Yellow Jacket Swarm Counter Unmanned Aerial System

Project Presentation

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Experience · Breadth of Impact

Yellow Jacket Space Program

Propulsion Engineer

LOX/Jet-A coaxial swirl injector · 92% c* efficiency · 12 hotfires

Georgia Tech Create-X

Co-Founder

Autonomous C-UAS swarm · DIU proposal filed · <$2K system

Deloitte

DoD/Intel Strategy

Defense & intelligence technology strategy consulting

Summit Capital

Founder / RE Dev

$30M LEED housing project · end-to-end ownership

Siemens Healthineers

R&D Engineer

Siemens Atellica · R&D simulation and testing

McKinsey / Home Depot

Design-to-Value / Product Dev

25% COGS reduction · workflow optimization for supply chains

SolidWorks · Fusion 360 · ANSYS Mechanical/Fluent · MATLAB · Python · C++ · CNC · 3D Printing · Waterjet

Project High Level Overview

Role:

- Designed and built the interceptor drone entirely. (Ideation, CAD, manufacturing, controls, testing)
- Now, lead 6-person team where I direct the hardware and manufacturing while integrating with SDEs

Timeline

- Last semester: Physical interceptor
- This semester: Drone Detection System
- Aim to reach TRL 6 by April

Program:

- Georgia Tech Create-X Team 7, Interstellar Foundry

Customers:

- DoW, DIU Open Solicitation, Army FUZE Program, SBIR/STTR

The Threat · Why Now

"Peace cannot be kept by force; it can only be achieved by understanding." — **Einstein**

Low-RCS, Low-Altitude

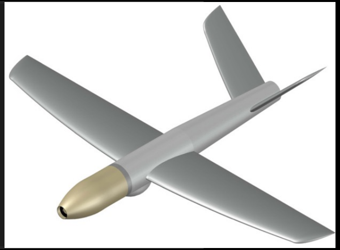
Commercial drones fly under legacy radar thresholds and Small cross-sections evade S-band sensors designed for aircraft.

Sustainment Cost

Current COTS counter-drone solutions cost $50K–$500K per node. Cannot scale to protect dispersed DoD infrastructure.

Fixed-Wing Gap

Fixed-wing interceptors lack agility to track quadcopters making abrupt evasive maneuvers at 25+ mph in close engagements.



OUR SOLUTION

Sub-$1,000 modular ground station

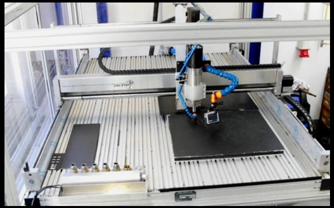
- NVIDIA Jetson Nano edge computing
- FMCW radar + CV sensor fusion
- Kalman filter → eliminates false positives
- 3× autonomous quadcopter interceptors
- Waterjet CF frames → < $280/unit
- < 15 min deployment on ISV/JLTV

Georgia Tech Create-X · DIU PROJ00656

UAV Interceptor · Manufacturing Approach

HARDWARE SPECS

Frame	450mm quad · waterjet-cut carbon fiber · 3D-printed mounts
Motors	2204 2300KV brushless x 4 per airframe
Props	5" balanced props · optimized for 80+ km/h intercept
Flight Ctrl	Matek F405 running ArduCopter firmware
Comms	ESP32 WiFi MAVLink telemetry to ground station
Cost	~$280/unit · ~$1,617 total for 3 swarm C-UAS
Weight	< 10 lb deployable ground station
Deploy	< 15 min setup



DESIGN FOR MANUFACTURABILITY

Waterjet-Cut CF Frames

2D DXF files → waterjet table. Minimal downtime. Consistent tolerance ±0.005"

3D-Printed Mounts & Arms

PETG/ABS mounts for motors, ESCs, sensors. Rapidly iterate design.

COTS Avionics Stack

Matek F405 + ESP3: proven, in-stock, programmable. No defense foundry lead times. ATO-streamlined.

Cost Scalability

Tier 1: ~$280/unit MVP.
Tier 2 (6-25): Low-Rate Production.
Tier 3 (26-100+): mass manufacturing.

Flight Physics · Design Rationale

PROPULSION PHYSICS

Hover condition:

4T = mg | T = (1/2)rho*A*v_i^2

T_total ~120 N, mass=0.280 kg -> T/W = 43.7

Drag at 80 km/h (22 m/s):

F_drag = (1/2)*rho*Cd*A*v^2

= (1/2)(1.2)(0.4)(0.04)(22)^2 = 4.7 N

Kinetic impact energy at intercept:

KE = (1/2)*m*v_rel^2

Head-on closing @ 44 m/s: KE ~ 271 J

Stress in CF frame (motor mount):

sigma_vm = sqrt(sigma^2 + 3*tau^2)

FEA: 48 MPa at mount holes, FS > 12

DESIGN CHOICES

450mm frame selection

Longer moment arm -> torque per RPM gain. Larger props -> efficiency. Tradeoff: agility vs. 250mm race frame. Intercept geometry prioritizes speed over maneuverability.

2204 2300KV motor selection

High KV -> high RPM at 3S -> fast prop speed -> agility. T/W >4 at hover. >75% thrust reserve for intercept acceleration from 0 to intercept < 3 sec.

3K twill CF plate (3mm)

sigma_UTS ~600 MPa. Waterjet-cuttable with +/-0.005" tolerance. Consistent cross-section vs. hand-layup. Safety factor >12 at motor mounts under crash load.

Polycarbonate guards over CF-PLA

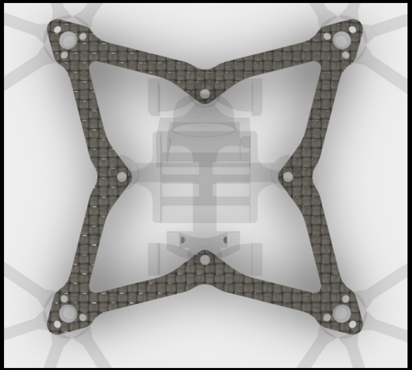
CF-PLA: 220N peak load but catastrophic brittle fracture. PC: 150N but plastic deformation. Failure MODE trumps failure LOAD for attritable interceptors. Must survive, not just endure.

Interceptor Design · CAD Modeling

Airframe: 450mm quadcopter configuration. Waterjet-cut carbon fiber main plate (3K twill weave, 3mm thickness). CAD modeled in Fusion 360. 3D-printed PETG motor mounts and camera bracket assembly.

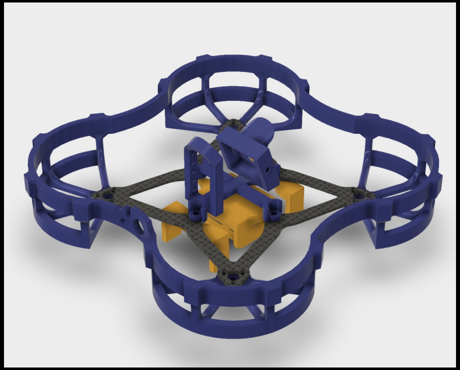
Propulsion: 2204 2300KV brushless motors × 4, 5" balanced propellers, optimized for 80+ km/h intercept.

Flight Control: Matek F405 running ArduCopter 4.3. ESP32 MAVLink telemetry bridge. Coordinated engagement vectors uplinked from ground station via Python API.

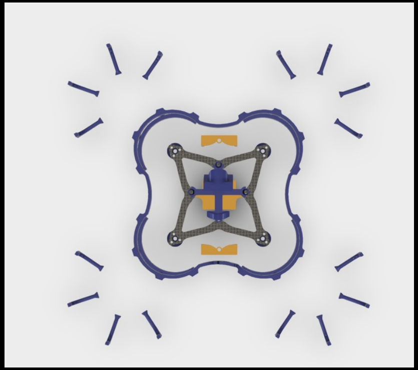


CF Frame (top view)
Waterjet DXF → waterjet table

Payload + Camera Mount
3D-printed PETG



Full Assembly: CF frame (gray) + guards (blue) + mounts (gold)



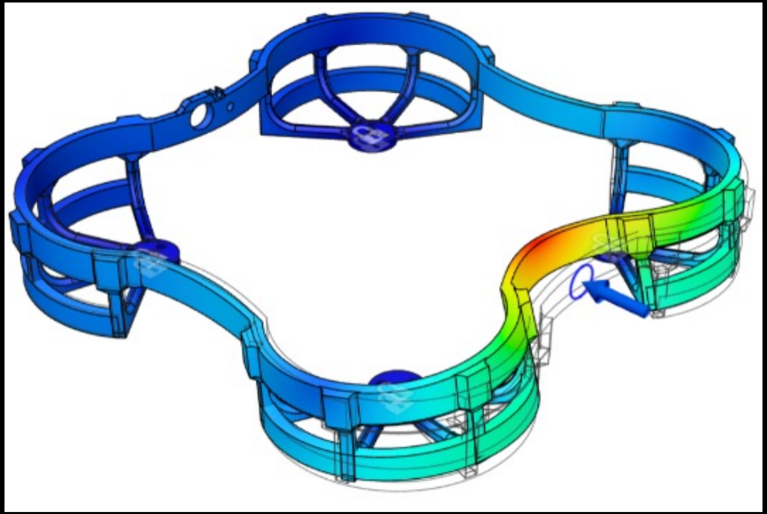
Full Assembly Exploded View

Material Selection · Structural FEA

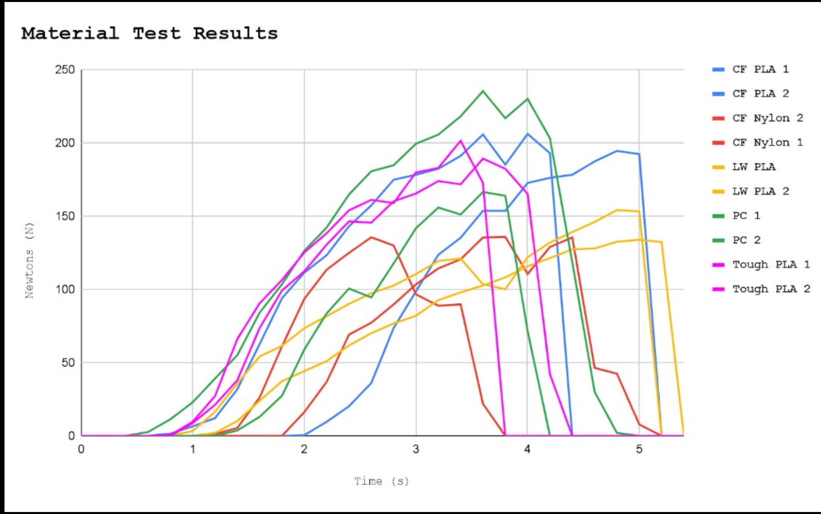
Test Campaign: 10 prop guard specimens across 5 filament types (PETG-CF, CF-PLA, CF-Nylon, LW-PLA, Polycarbonate) loaded to failure under axial compression. Peak load plotted over time to characterize both maximum strength and failure mode (brittle fracture vs. progressive deformation).

Key Finding: CF-PLA achieved peak loads exceeding 220N but fractured catastrophically. Polycarbonate showed the best post-failure behavior (plastic deformation rather than shattering), selected as primary material for guards.

Von Mises Stress: $\sigma_{vm} = \sqrt{\sigma_\theta^2+\sigma_{ax}^2-\sigma_\theta \cdot \sigma_{ax}}$



ANSYS FEA: Prop guard structural analysis
Von Mises stress distribution under axial compression load.



Material test result: Force (N) vs. time (s) across 10 specimens.

Physical Build Setup

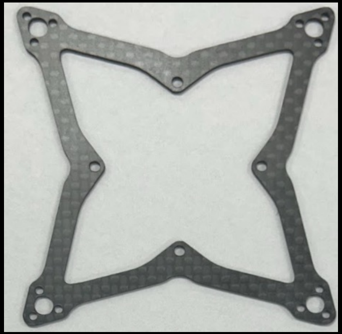
Carbon Fiber Main Plate: 2D DXF exported from Fusion 360 and cut on a waterjet table (±0.005" tolerance).

- **Challenge:** delamination with carbon fiber (piercing in scrap material, using lower pressure)

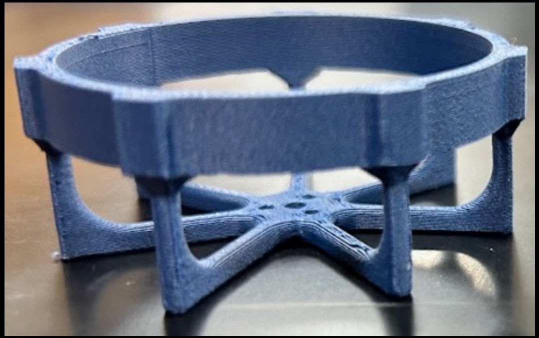
3D-Printed Components: Motor mounts, prop guards, and camera brackets printed in PETG and polycarbonate using Bambu Lab X1 Carbon. Iterated across 5 revs to achieve target budget and impact resilience.

- **Challenge:** Difficulty with polycarbonate (requires high enclosure temps to prevent warping)

Assembly Budget: ~$280/interceptor. Total 3-unit CUAS: ~$1,617.



Waterjet CF frame (physical part)



3D-printed motor guards

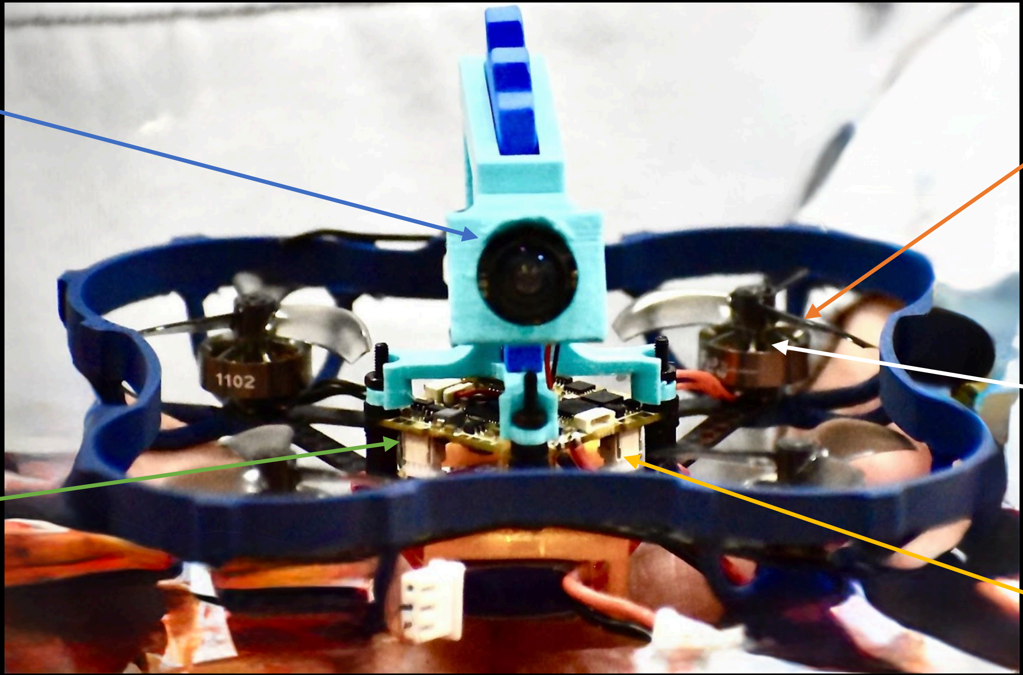
3D-printed prop guards

Camera/Payload mount finished part

Final Assembly · COTS Parts

Bee Eye

BLV4

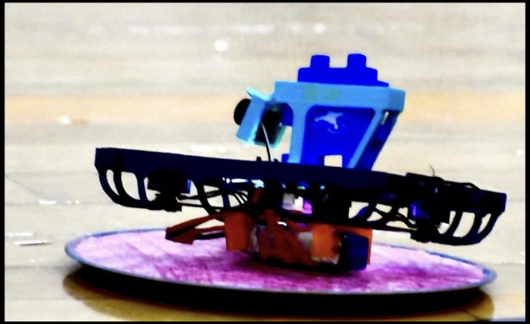


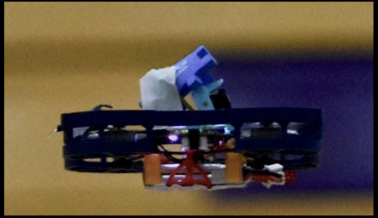
Flow 1102

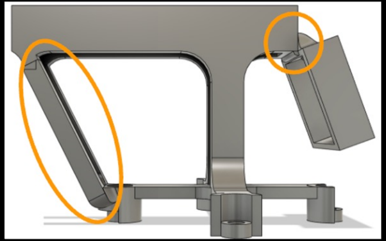
Azi Triblade

XT-30

Initial Flight Testing






Upon Testing, realized the camera mount was not strong enough, so decided to add structural supports to the camera mount itself.

Failures · Adjustments

Camera mount collapsed in flight

Vibration + thrust reversal loads not in single post design.

PID oscillations / unstable hover

ArduCopter defaults tuned for large/stiff frames; sub-300g frame has different inertia tensor.

Prop guard CF-PLA: brittle fracture in crash

CF-PLA 220N peak but catastrophic shatter; debris hazard, system unflyable.

Camera FOV missed target in terminal intercept phase

Level-mounted camera; approach from above geometry requires forward down FOV.

3-point triangulated PETG mount, silicone damping grommets.
Design for dynamic loads, not static geometry. Verify with first principles

Tuned PIDs from scratch; notch filter at 200Hz; Blackbox logging.
Never assume firmware defaults work. Tune methodically.

Tested 5 materials; selected PC (150N, plastic deformation, re-flyable).
Failure MODE > peak strength.
Ductile failure > Tensile Strength

-15 deg angled mount; recalibrated YOLO bounding box coordinates
Model engagement geometry in 3D first. Kinematic analysis before hardware.

Detection System Architecture

01

Primary Detection

RADAR

24 GHz FMCW
K-band Doppler
Range + velocity
≤20 dBm emission

→

02

Secondary Tracking

OPTICAL

OAK-D Pro stereo
Hikvision 8MP
Angular precision
Object classification

→

03

Edge Compute

JETSON

NVIDIA Jetson Nano
60+ TOPS, 7-25W
YOLOv8 inference
No cloud dependency

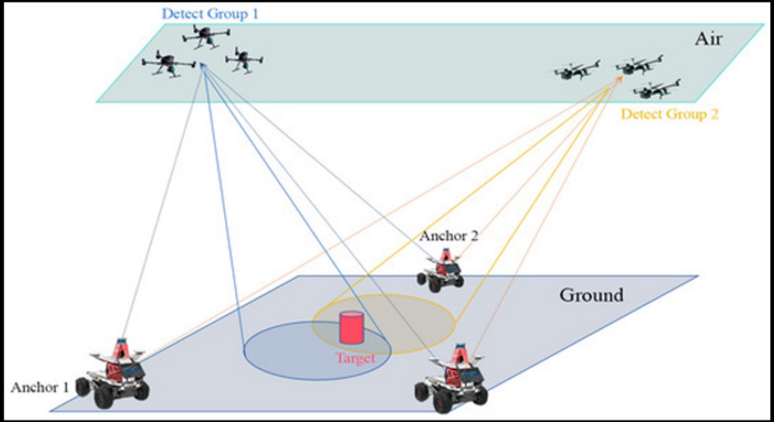
→

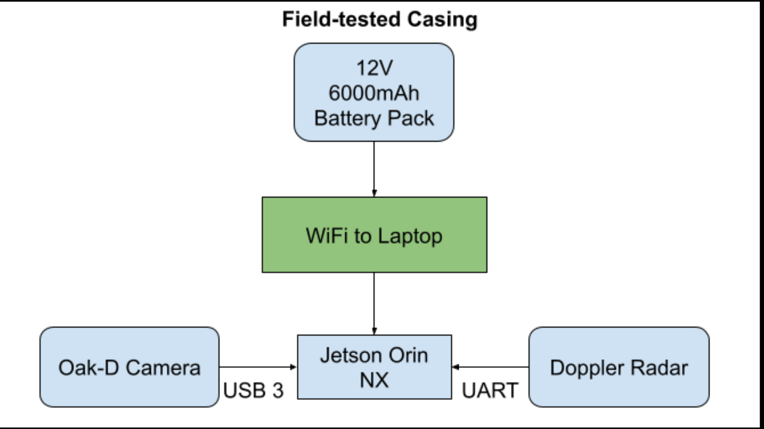
04

Sensor Fusion

HANDSHAKE

Kalman filter
Fuses radar + optics
Elim. false positives
Locks Group 1/2 sUAS





Sensor Fusion Physics · Kalman Filter

The math that locks Group 1/2 targets while eliminating birds and weather clutter

RADAR DETECTION

Doppler frequency shift:

$$f_D = 2v_r \cdot f_c / c$$

Radar range equation:

$$R = [(P_t \cdot G_t \cdot G_r \cdot \lambda^2 \cdot \sigma) / ((4\pi)^3 \cdot S_{min})]^{(1/4)}$$

FMCW range resolution:

$$\Delta R = c / (2 \cdot B) \quad \text{where } B = \text{bandwidth}$$

Velocity resolution:

$$\Delta v = \lambda / (2 \cdot T_{CPI})$$

$\boldsymbol{f_D}$: Doppler frequency shift, $\boldsymbol{v_r}$: Relative velocity, $\boldsymbol{f_c}$: Carrier frequency,
$\boldsymbol{c}$: Speed of light, $\boldsymbol{R}$: Target range, $\boldsymbol{P_t}$: Transmit power, $\boldsymbol{G_t}$: Transmit antenna gain,
$\boldsymbol{G_r}$: Receive antenna gain, $\boldsymbol{\lambda}$: Wavelength, $\boldsymbol{\sigma}$: Radar cross section,
$\boldsymbol{S_{min}}$: Minimum detectable signal, $\boldsymbol{\Delta R}$: Range resolution, $\boldsymbol{B}$: Bandwidth,
$\boldsymbol{\Delta v}$: Velocity resolution, $\boldsymbol{T_{CPI}}$: Coherent processing interval

360° Azimuth Coverage

10 Hz Update Rate

KALMAN FILTER FUSION

State prediction:

$$\hat{x}_{k|k-1} = F \cdot \hat{x}_{k-1} + B \cdot u_k$$

Covariance prediction:

$$P_{k|k-1} = F \cdot P_{k-1} \cdot F^T + Q$$

Kalman gain:

$$K_k = P_{k|k-1} \cdot H^T \cdot (H \cdot P_{k|k-1} \cdot H^T + R)^{-1}$$

State update:

$$\hat{x}_k = \hat{x}_{k|k-1} + K_k \cdot (z_k - H \cdot \hat{x}_{k|k-1})$$

$\widehat{x_{k|k-1}}$: Predicted state estimate, $\boldsymbol{F}$: State transition model, $\boldsymbol{z_k}$: Measurement vector
$\widehat{x_{k-1}}$:: Previous state estimate, $\boldsymbol{B}$: Control input model, $\boldsymbol{u_k}$: Control vector,
$\boldsymbol{P_{k|k-1}}$: Predicted covariance estimate, $\boldsymbol{Q}$: Process noise covariance, $\boldsymbol{K_k}$: Kalman gain,
$\boldsymbol{H}$: Observation model, $\boldsymbol{R}$: Measurement noise covariance, $\widehat{x_k}$: Updated state estimate

<5 mrad Az/El Error

50 mph Vehicle Track Speed

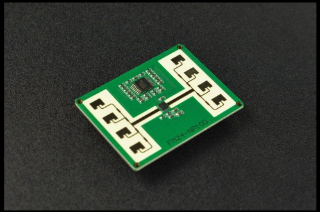
Hardware Subsystems

Radar (Tiered Scaling) ⟶ **Optical Suite** ⟶ **Edge Processing** ⟶ **Tactical Footprint**

Radar (Tiered Scaling)

↓

mmWave Microwave Radar (~$66) for MVP and >30m tracking.



Optical Suite

↓

Highly compatible with **OAK-D Pro** (RGB/stereo depth) or

Edge Processing

↓

NVIDIA Jetson Nano delivers localized, cloud-independent AI (60+ TOPS) on a highly efficient **7-25W power draw.**

Tactical Footprint

↓

Sub-10 lb modular design supports under 15-minute deployment on ISV, JLTV, FMTV, and HEMTT platforms.

Software Subsystems

GNSS-Denied Navigation ⟶ Spectral Stealth ⟶ Cybersecurity

GNSS-Denied Navigation

↓

Converts 2D pixels to 3D map coordinates using OpenCV and SLAM to localize threats without GPS.

Spectral Stealth

↓

Air-gapped serial communication and passive optics ensure maximum EMI robustness and low physical/spectral signatures.

Cybersecurity

↓

Architecture strictly complies with **DOD Instruction 8510.01 (RMF)** to streamline the Authorization to Operate (ATO).

LOE 1 Homeland ASCII output · FAAD C2 · 5-30kW gen compatible

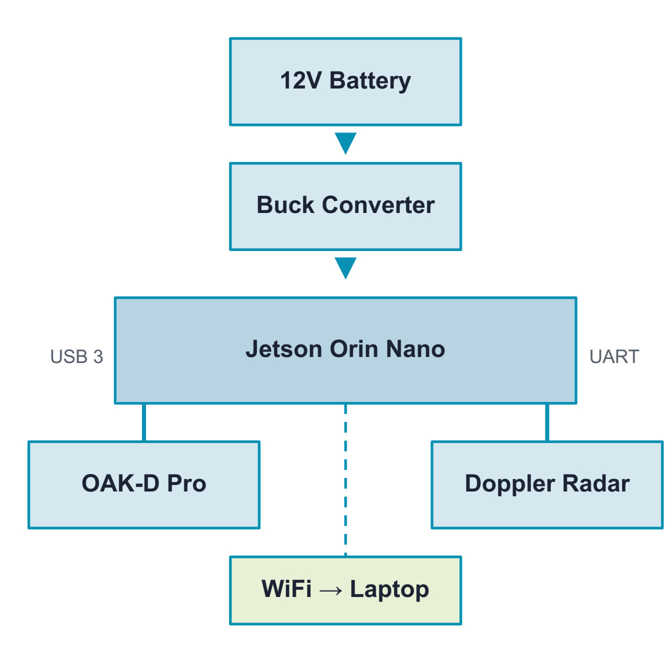
LOE 2 Tactical/Mobile GNSS-denied · ISV/JLTV mount · passive posture

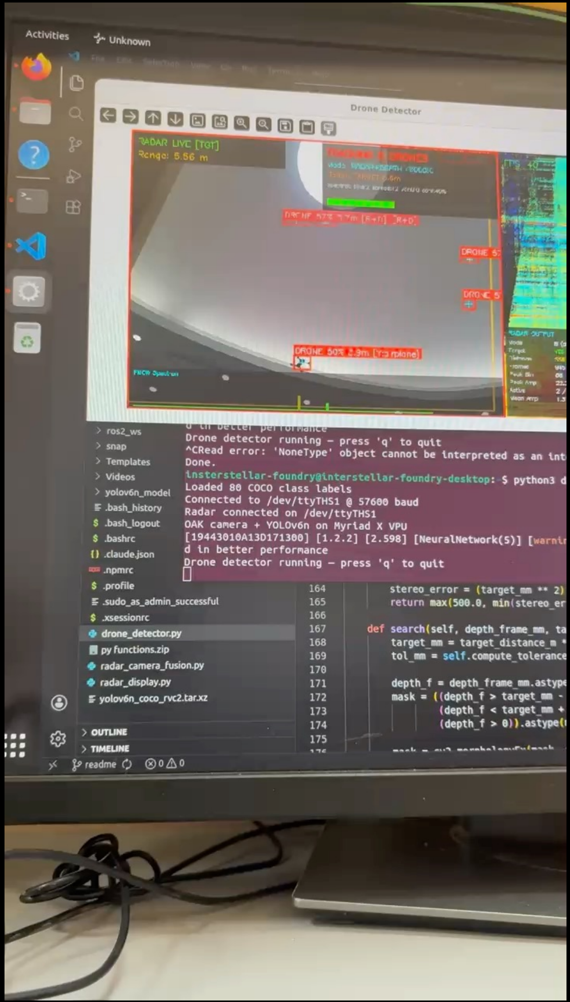
BILL OF MATERIALS

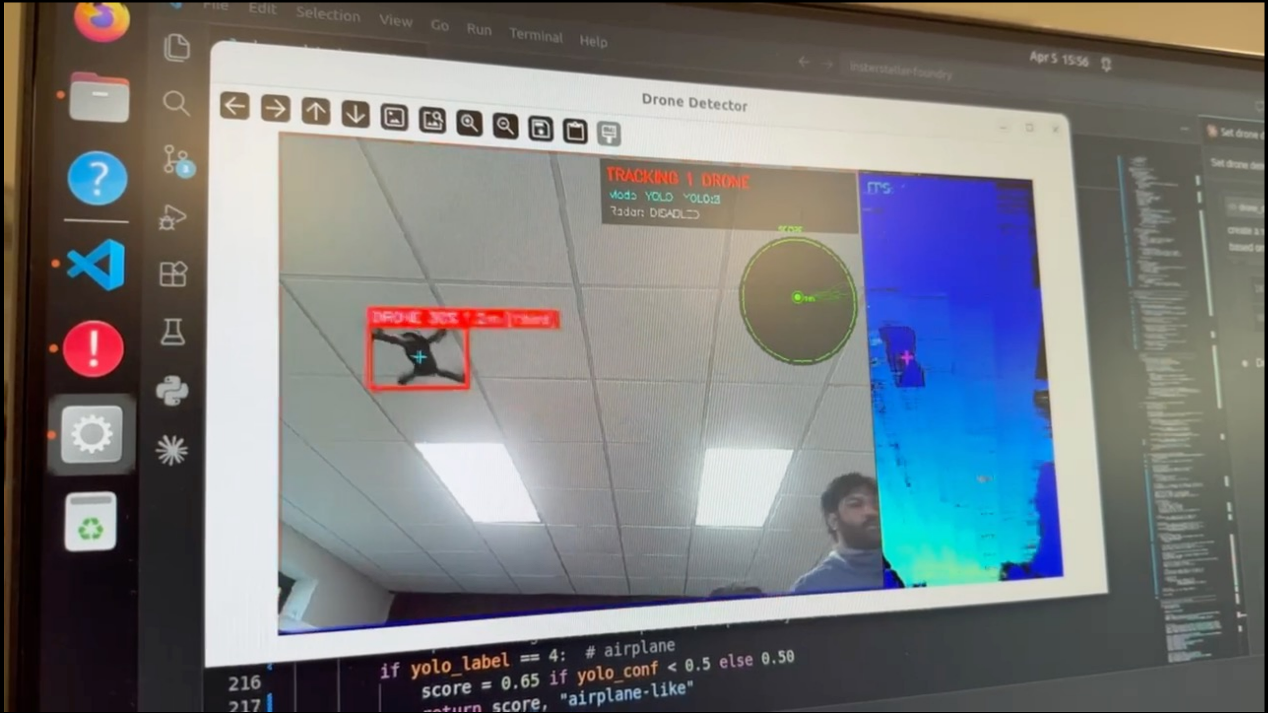
Component	Supplier	Cost
OAK-D Pro Camera	Neobits	$302.19
Jetson Orin Nano	Amazon	$245.00
12V 6000mAh Battery	Fat Tire House	$65.00
mmWave 24GHz Radar	DFRobot	$65.90
DC Converter + Cables	Amazon	$27.07
Test Drones	Amazon	$39.98
Sandisk MicroSD	Amazon	$56.40
Tax + AI Tools	Various	$175
TOTAL		**$957.15**

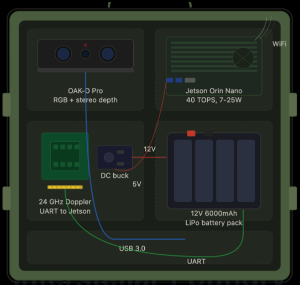
Left to purchase: Pelican Case <$100

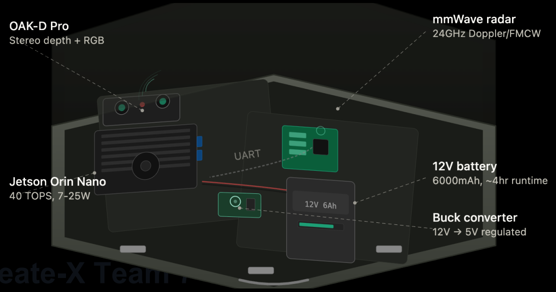
Hardware Schematic

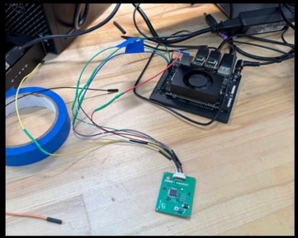












Interstellar Foundry • Create-X Team

Engagement Logic · Swarm Software

GROUND STATION SOFTWARE

Python / OpenCV

Real-time video processing pipeline

HB100 Doppler

Velocity + range primary detection

Kalman (FilterPy)

Multi-target tracking + fusion

YOLOv8 inference

Jetson-accelerated classification

PyQt6 UI

Operator display / engagement cue

ENGAGEMENT LOGIC

Target acquired

Dual-sensor confirm → lock

Primary dispatch

Lead interceptor → head-on vector

Flankers launch

2 flankers on diverging intercept

Abort logic

Flankers abort if primary scores kill

Re-engage

Missed intercept → re-cue loop

MVP SUCCESS CRITERIA

Dual-sensor detect

Both radar AND camera confirm

80+ km/h targets

Intercept at full speed

3 simultaneous launches

Coordinated swarm execution

Kinetic contact

Physical intercept confirmed

< 15 min deploy

From packed to combat ready

Business Case · Current Market

The gap: low-cost, high-lethal, mass-producible C-UAS at scale

TIERED ROM PRICING

TIER 1 — 1-5 units

$500-$1,000/node

Prototype / evaluation

TIER 2 — 6-25 units

LRIP pricing

Low-rate initial production

TIER 3 — 26-100+ units

Mass manufacturing

Full contract scale

MARKET CONTEXT

$4B+

C-UAS global market by 2030

~$500K

Avg. legacy sensor node cost

<$1K

Our node cost is 500× cheaper

Group 1/2

Primary threat is underserved

Other Projects

LOX/Jet-A Coaxial Swirl Injector

Yellow Jacket Space Program · Propulsion Engineer

SITUATION

YJSP's prior pintle injector failed after 5 seconds due to thermal melting. Vespula flight vehicle needed 35-second burn to reach 100,000 ft. A more robust injector was required.

ACTION

Waterflow campaign (8 tests) to validate Cd models. 10 subscale hotfires (7-element, 261 psia, MR=1.8). 1D thermal resistance model + Ansys FEA. LN2 cryogenic validation. Mfg gap fix (gasket + machined ring).

TASK

Design, analyze, and test a LOX/Jet-A coaxial swirl injector achieving ≥85% c*-efficiency while maintaining thermal structural integrity throughout a full-duration flight burn.

RESULT

92% c*-efficiency: highest in YJSP history. Zero melting after 10 hotfires including 8-second burn. Massive improvement over previous 5-second failure limit.

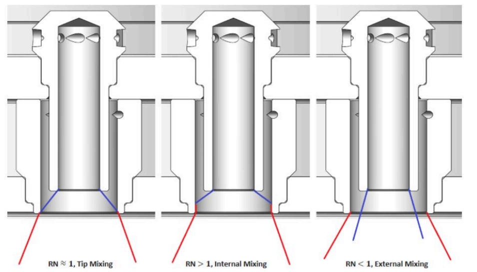


Fig 1. Mixing location

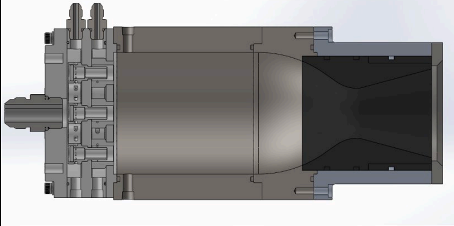


Fig. 4. Subscale combustor CAD cross section

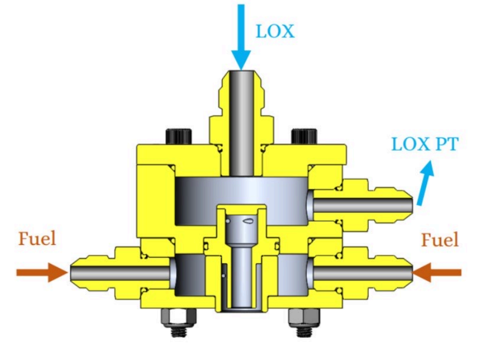


Fig. 2. Single element swirler assembly.

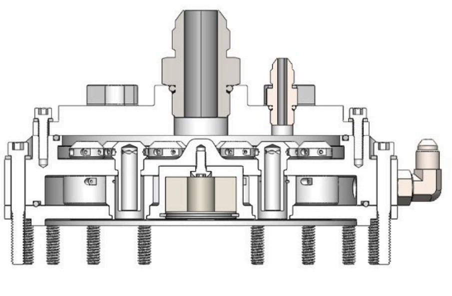


Fig. 12. Vespula injector CAD cross section

Responsible Engineer for Injector Design and Machining

MY ROLE

Research & Concept
- Studied published NASA research and models
- Proposed coaxial swirl to team.
- Designed full test campaign.

Machining
- CNC/Lathe: outer manifold bodies, swirler nozzles
- Milling: entry holes (0.06”) manifold faces
- Total: 6 manifold assemblies, 54 swirler elements.

Team delegation
- ANSYS FEA thermal validation: delegated to 2 senior team members (new to me at the time)
- Hot fire test operations: team of 5.
- Waterflow data processing: shared.
- I owned design, machining, and iterating.

What went wrong
- 0.005" machining gap -> fuel Cd doubled -> 14 psi drop vs. 58 psi predicted.
- Root cause: tolerance miscalculation in manifold/swirler interface.
- Fix: gasket + machined ring on outer fuel swirler.

CNC OPERATIONS · MATERIAL RATIONALE

Material: 303 Stainless Steel
- Selected over 304/316: free-machining sulfur addition reduces work hardening.
- 3 flute cobalt drill, 600 RPM, flood coolant.
- Ra 32 on all sealing faces.

Turning operations
- Swirler nozzle OD: +0.000/-0.001" for press fit into manifold.
- Nozzle bore: +0.001"/+0.000" (controls Cd)

Milling operations
- Entry holes: Per element, drilled tangentially.
- Boss port tooling: Carbide AS5202-12 LOX inlet.
- Port locations for PT/TC: +/-0.002" tolerance.

Scale path: 7-element -> 18-element
- More elements -> finer atomization -> c* up.
- Pressure drop scales with element count.
- Subscale 7-element: 83% c*. 18-element: 92%.

Vacuum furnace brazing (outsourced)
- 18 swirlers brazed simultaneously at 1100C.
- Gap post-braze was the key failure point.
- Next version: add tolerance analysis to pre-braze.

Injector Subscale · Thermal Analysis

DISCHARGE COEFFICIENT & SPRAY ANGLE

Discharge coefficient:

$$Cd = \dot{m} \,/\, (A\sqrt{(2\rho\Delta p)})$$

Fuel Cd (Vespula flight):

$$Cd = 0.0194 \quad | \quad \Delta P = 90\ \text{psi} \quad (30\%\ \text{stiff})$$

c*-efficiency:

$$c^* = pc \cdot At \,/\, \dot{m} \quad (\text{vs. NASA CEA theoretical})$$

1D THERMAL RESISTANCE + FEA VALIDATION

Total thermal resistance:

$$R\_total = 1/(hc \cdot A) + t/(k \cdot A) + 1/(hg \cdot A)$$

Heat flux:

$$q = (Tg - Tc) \,/\, R\_total$$

1D predicted face temp (hot side):

$$T = 1{,}194\ K \quad (1D) \quad \text{vs} \quad 1{,}365\ K \quad (FEA)$$

Fig. 5. Subscale injector spray visualization

Fig. 9. LN2 injector cryo flow

Fig. 6. Subscale engine hotfire picture

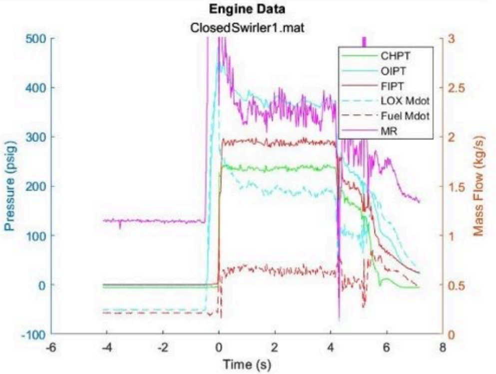


Fig. 7. Subscale engine hotfire data

Full Scale · Hot fire Testing

RESULT: 18-element Vespula flight injector → 92% c*-efficiency (highest in YJSP history)
Zero melting after 10 hot fires incl. 8-second burn

Fig. 10. LOX posts after hotfire

Fig. 11. Injector faceplate after hotfire

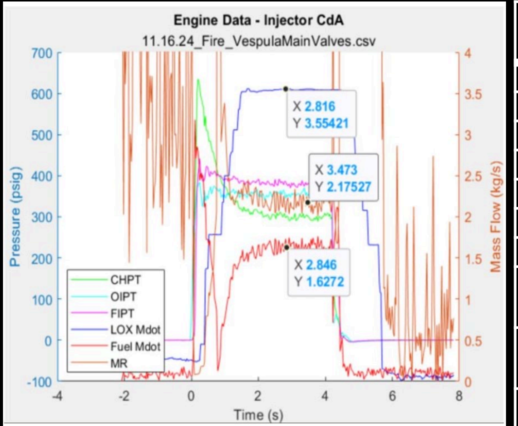


Fig. 16. Vespula injector hotfire data.

Hotfire Number	Calculated c*-efficiency (%)
1 (Open LOX)	75%
2 (Open LOX)	75%
3 (Closed LOX)	70%
4 (Open LOX)	78%
5 (Open LOX)	79%
6 (Open LOX)	82%
7 (Open LOX)	85%
8 (Open LOX, internal mixing)	87%
9 (Closed LOX, internal mixing)	74%
10 (Open LOX, internal mixing)	92%

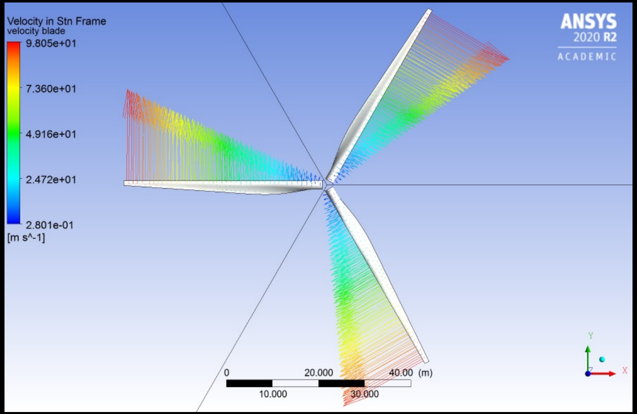
Colorado Wind Farm · Turbine Design

AE 4701-A Wind Engineering · WT_PERF + ANSYS Fluent/Mechanical

TURBINE SPECIFICATIONS

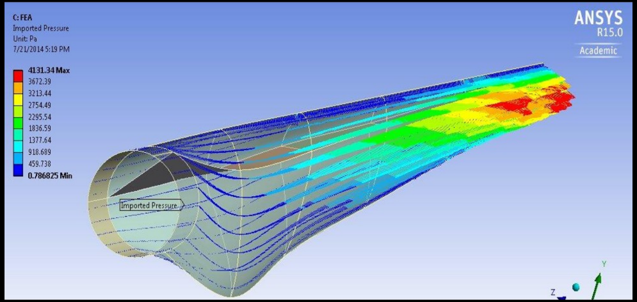
Rating	1 MW (1,000 kW)
Blades	3 (industry standard)
Rotor radius	15 m (30 m diameter)
Hub height	30 m
Hub radius	0.75 m
Precone	2° (reduces gravity bending moment)
Shaft tilt	5° (reduces loads)
Air density	ρ = 1.08 kg/m³ (1,600 m altitude)
Optimal pitch	4° (best power across all wind speeds)

BLADE VELOCITY



Velocity in Station Frame (blade ref.): 0.28 to 98.1 m/s. Tip velocity peaks at outer radius due to ω·r. Three-blade rotor, 30 m diameter. Radially varying velocity confirms blade element theory loading distribution.

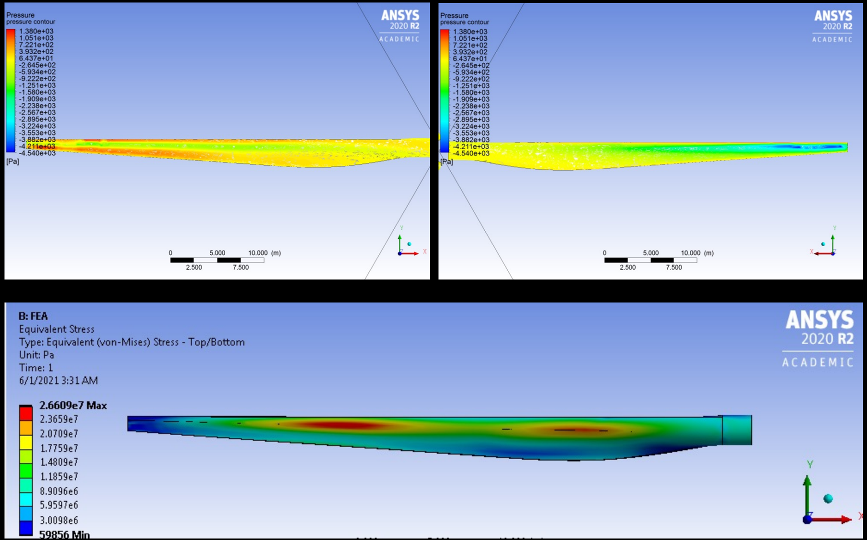
CFD · IMPORTED PRESSURE DISTRIBUTION



Aerodynamic pressure mapped from Fluent onto blade surface. Max pressure 4,131 Pa at tip (outboard 30%). Baseline near root. Used as boundary condition for structural FEA.

Wind Turbine Blade · CFD/FEA Results

FEA · VON MISES STRESS (Front/Back)



PERFORMANCE & ECONOMICS

758 MWh

Annual energy capture @ 4 m/s avg wind

8.66%

Capacity factor (Weibull k=2.0)

74.7%

Energy capture ratio vs. Betz

$0.50/kWh

Lowest cost source in Colorado

$495K

Installed cost/kW (1 MW baseline)

CFD Pressure (Fluent + FSI): Aerodynamic pressure mapped from Fluent onto blade surface. Max pressure 4,131 Pa at tip (outboard 30%). Baseline near root. Used as boundary condition for structural FEA.

FEA Structural (Mechanical): Von Mises stress max: 26.6 MPa at root/mid-chord region. Blade span 15 m. Stress concentrations identify critical spar locations for material and thickness optimization.

Aero-Structural Insight: Tip loading drives structural demand. Root section carries bending and torsion from lift distribution. Blad tapering, twist distribution, and material selection for NREL-class blades.

Lab for Ultrasonic Non-Destructive Evaluation

Georgia Tech · GT-CNRS Metz, France · Dr. Nico Declercq

PROBLEM

Ship hull inspection requires robots to localize on metal without GPS. Traditional dry-dock cycles take 8 days and cost millions.

APPROACH

Monte-Carlo Localization via particle filter. Two co-located piezos act as pulse-echo sonar. Lamb wave echoes from plate boundaries encode robot position.

SETUP

60×45 cm Al plate (6mm). 108 positions, 216-step path. 500 particles. 100 kHz / 100V / 1.25 MHz.

KEY RESULTS

<1 cm

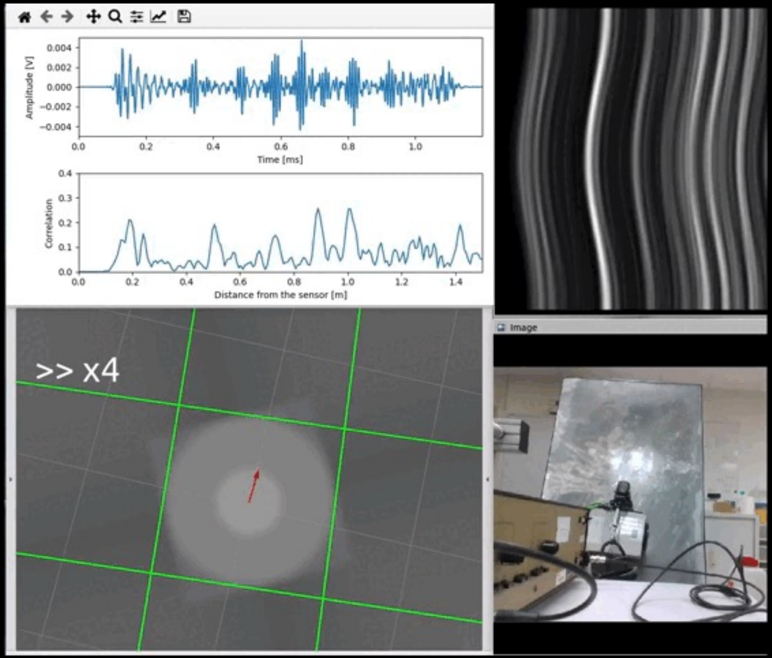
Localization accuracy over full 216-step trajectory

12 steps

Particle convergence from 1/4-plate initialization

DEFENSE RELEVANCE

GPS-denied localization on metallic platforms. Particle filter logic mirrors Kalman sensor fusion in C-UAS swarm. SLAM on unknown hull geometry.



Crawler traversing aluminum plate: particle filter localizes in real time from Lamb wave echoes to plate boundaries

Echo Detection · Signal Processing

Propagation model · Cross-correlation · Hilbert envelope · Probabilistic particle update

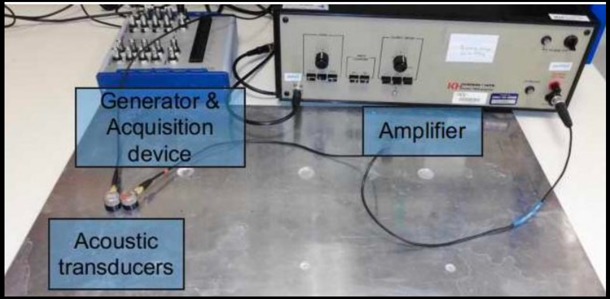
ECHO DETECTION PIPELINE

1 Propagation Model
Predict echo arrival times from geometry using A0-mode Lamb wave velocity

2 Cross-Correlation
Correlate received waveform with transmitted burst to identify echo peaks

3 Hilbert Envelope
Analytic signal envelope extracts time-of-flight for each first-order boundary

4 Probabilistic Update
Exponential likelihood: $p(z|x) = \exp(-k \cdot |d_{meas} - d_{pred}|)$. Re-sampling update



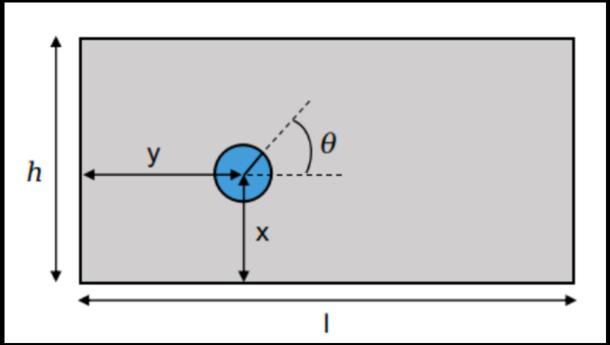
HARDWARE & PHYSICS

Transducer Setup
2 co-located piezos (exciter + receiver). 100 Vpp. 100 kHz burst. 1.25 MHz.

Lamb Wave Physics
A0 antisymmetric mode dominant at 100 kHz in 6mm Al. Dispersive: group velocity varies with frequency. Model calibrated to measured dispersion curve.

Limitations
Dual-transducer approximation has bearing error. Only First-order reflections.

Future Work
Single EMAT transducer (non-contact). SLAM for unknown geometry. Hull testing.



Thank you!

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